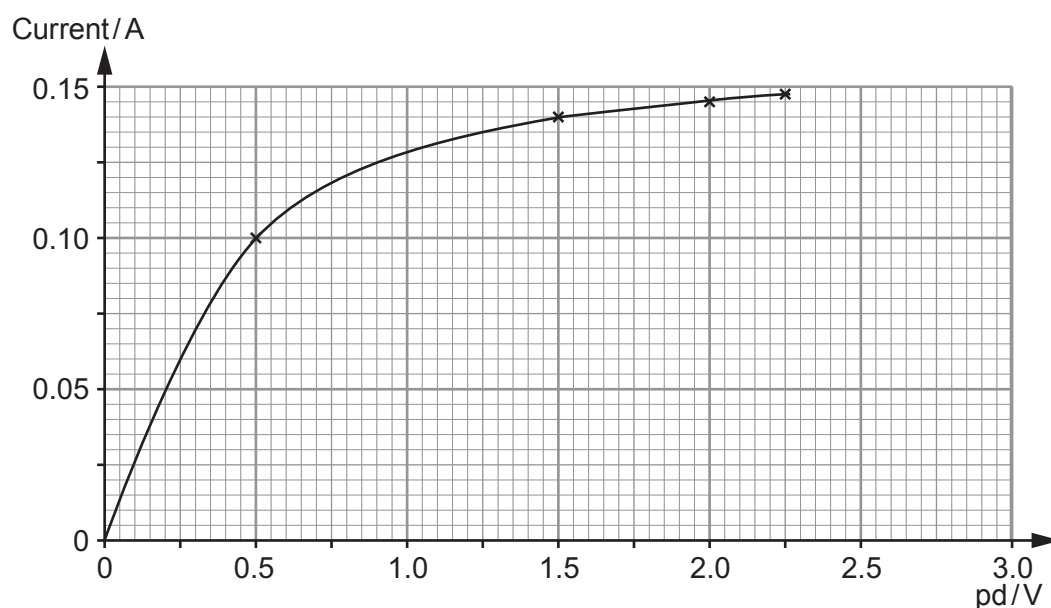


7. (a) (i) Draw the circuit diagram for an investigation of how the current through a filament lamp varies with the applied potential difference. [2]

- (ii) The lamp is labelled “3V, 0.16A”. The ammeter to be used is a multimeter with a 0 – 200 mA range and a 0 – 10A range. State which range should be selected, and justify your choice. [1]

- (iii) The lamp has already been investigated by a student, Sion, who plotted the graph reproduced below. State **two** ways in which his investigation (**not** his graph plotting) could have been improved. [2]



- (b) (i) Use the graph to calculate the ratio: $\frac{\text{Resistance of lamp at 2.00 V}}{\text{Resistance of lamp at 0.50 V}}$ [3]

Examiner
only

- (ii) Explain, in terms of free electrons, why the temperature **and** the resistance of a metal wire increase when the potential difference across the wire is increased. [6 QER]

**TURN OVER FOR THE LAST
PART OF THE QUESTION**



- (c) A scientist claims to have made a material which is superconducting at room temperature. He publishes his procedure. Several research teams try to follow the same procedure, but without success. Discuss the arguments for and against spending more public money following up the scientist's claim. [3]

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END OF PAPER

